Problem 12.3

An investor makes three deposits into a fund, at the end of 1, 3, and 5 years. The amount of the deposit at time t is $100(1.025)^t$. Find the size of the fund at the end of 7 years, if the nominal rate of discount convertible quarterly is $\frac{4}{41}$.

Solution

The quarterly discount rate is

$$\frac{1}{4}\left(\frac{4}{41}\right) = \frac{1}{41},$$

so the accumulation factor per quarter is

$$\left(1 - \frac{1}{41}\right)^{-1} = \frac{41}{40} = 1.025.$$

Thus, the annual accumulation factor is $(1.025)^4$.

Each deposit accumulates from its deposit date until the end of year 7. Therefore, the fund value at year 7 is

$$\begin{aligned} F_7 &= \sum_{t\in\{1,3,5\}} 100(1.025)^t \left((1.025)^4\right)^{7-t} \\ &= 100(1.025)^1 \left((1.025)^4\right)^6 + 100(1.025)^3 \left((1.025)^4\right)^4 \\ &\quad + 100(1.025)^5 \left((1.025)^4\right)^2 \\ &\approx 483.110533. \end{aligned}$$

Hence, the size of the fund at the end of 7 years is

$$\boxed{483.11}.$$